Motorized Tape Springs as Prismatic Links in Deployable Parallel Mechanisms for Space Exploration Robots

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Background and Research Gap

Planetary exploration requires robots with long reach and compact stowage, creating a trade-off between stowage volume and workspace. Serial robots offer large workspaces, but require higher stowage volume. Parallel mechanisms provide higher stiffness, better load distribution, and compact actuator placement, but offer limited workspace (Antonov, 2024).

Tape springs can address this trade-off by rolling into compact coils and deploying as structural members (Tang et al., 2020; Ding et al., 2022). Recent studies demonstrate their use in long-reach, climbing, and extensible-bending robotic limbs (He et al., 2025; Morton et al., 2024; Quan & Hong, 2022; Quan et al., 2023). However, current research predominantly focuses on isolated, single-limb applications. Their integration into a parallel mechanism for mobility and manipulation remains underexplored. Tape-spring-based prismatic links in parallel mechanisms can potentially achieve compact stowage and extended reach.

Hypothesis

Motorized tape springs can replace the rigid prismatic links of a parallel mechanism, letting it extend far beyond its stowed size while still carrying and orienting a payload.

Prototyping and Testing

Prototype design. To test the hypothesis, a 3-RPS mechanism was developed using motorized tape springs as prismatic links. Three identical limbs join the base and the moving platform through revolute and ball joints (Fig. 1). Each limb has a tape blade, idle pulley, and motor-driven drum. As in EEMMMa-1 (Quan & Hong, 2022), the tape is folded into a U-shape over the pulley, so the two sections bend in opposite directions, increasing the stiffness and peak moment capacity. One tape end is fixed to the body, the other is wound onto the drum, and an elastic band keeps the tape taut. Turning the drum moves the pulley hub, providing prismatic motion. The parts are FDM-printed in PLA+ with MG945 servos. Off-the-shelf bearings and fasteners are used for assembly, and the tape blade is 16 mm commercial measuring tape. The total mass of the prototype is 627 g.

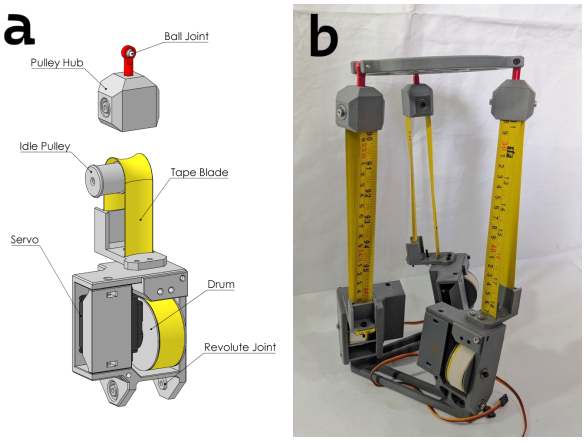


Fig. 1. The 3-RPS mechanism. (a) CAD view of one motorized tape-spring limb. (b) Assembled prototype.

Test method. The prototype was tested at effective tape-spring lengths of 10, 30, and 45 cm, with the platform level (0° inclination) and tilted to 20°. The inclination was limited to 20° to remain within the ±20° angular range of the ball joints and avoid prestraining the tape springs. At each length, the tape springs were first extended to the set lengths, and the load was then applied to the platform. For the inclined tests, one limb was extended further to tilt the platform.

Loads of 0.5 kg and 1 kg were planned. Failure was defined as buckling of the tape springs. At 10 cm, both loads were tested. At 30 cm, the 1 kg load at 0° inclination caused failure, so the remaining tests at 30 cm and 45 cm used only the 0.5 kg load.

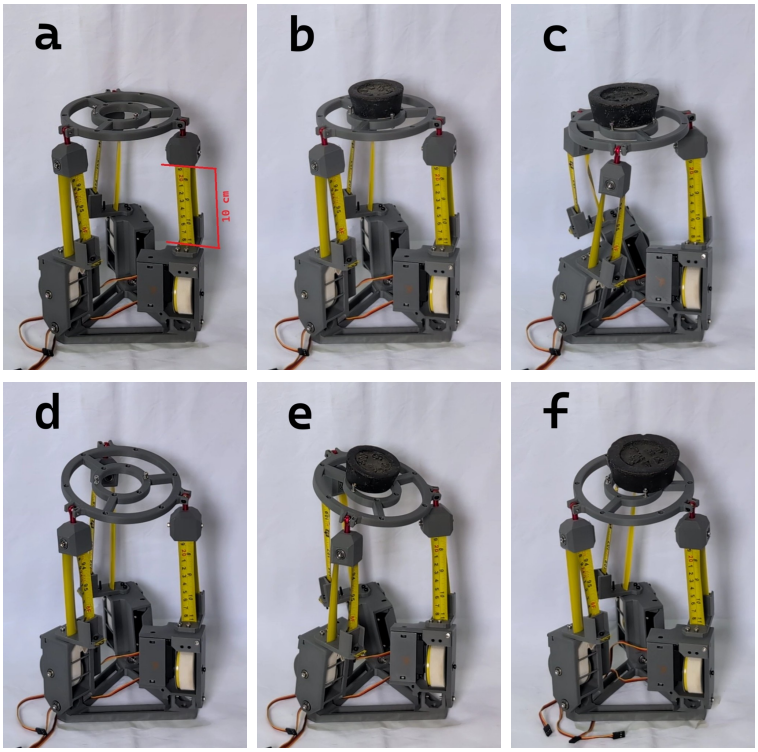


Fig. 2. Tests at 10 cm tape length. (a–c) Level platform; (d–f) platform tilted 20°. Each row: unloaded, 0.5 kg, 1 kg.

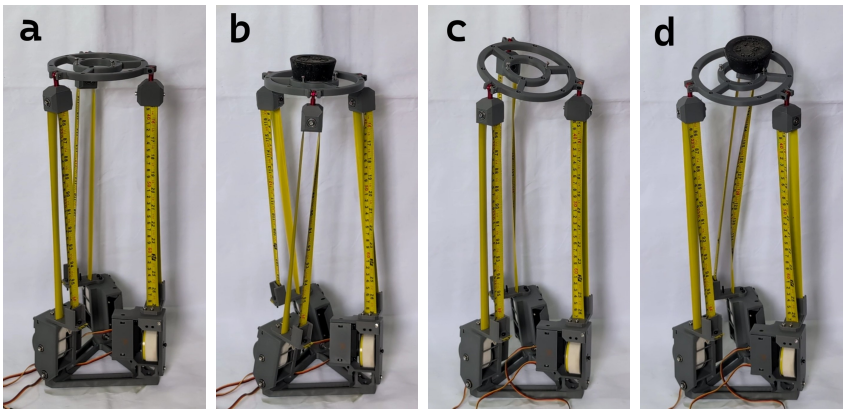


Fig. 3. Tests at 30 cm tape length. (a, b) Level platform: unloaded, 0.5 kg. (c, d) Platform tilted 20°: unloaded, 0.5 kg.

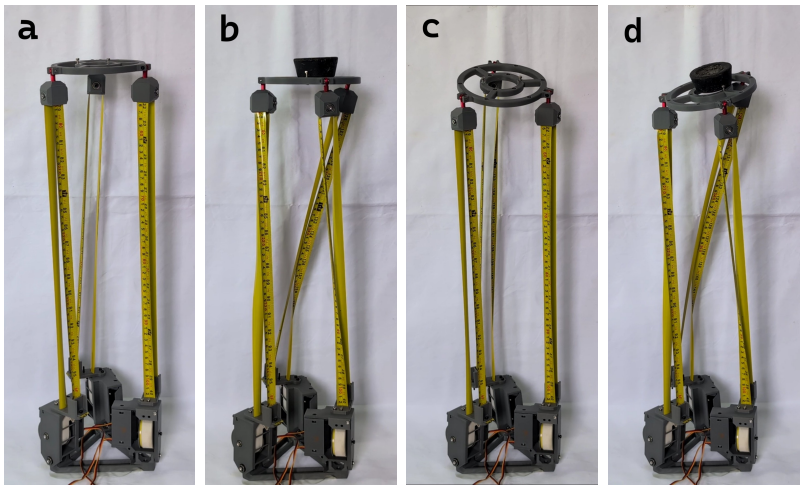


Fig. 4. Tests at 45 cm tape length. (a, b) Level platform: unloaded, 0.5 kg. (c, d) Platform tilted 20°: unloaded, 0.5 kg.

Discussion

The prototype demonstrates that tape springs can function as prismatic links in a parallel mechanism while revealing the limits of the design. At 10 cm extension, the mechanism remained stable under loads up to 1 kg, with only minor compliance at 20° inclination (Fig. 2). At 30 cm, it supported 0.5 kg at both orientations, but the 1 kg load caused tape-spring failure (Fig. 3). At 45 cm, it supported 0.5 kg at 20° inclination, but the blades underwent substantial bending and twisting (Fig. 4). Although parasitic yaw persisted throughout testing, higher extensions and loads substantially magnified this torsional effect.

As a proof of concept, the prototype does not represent the overall capability of the approach, but shows that motorized tape springs can provide variable-length links in a parallel architecture. Further research should investigate engineered tape springs, improved joints, and alternative parallel configurations to increase reach, stability, and mobility with low stowage volume requirements.

References

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